

Solution Requirements

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>No login required; application runs as a single, session-based tool accessed directly by the user.</i>	Low
2	Authorization levels	<i>No user roles or access levels; single-session usage per browser tab, no differentiation between users</i>	Low
3	External interfaces	<i>OpenAI Whisper, Sentence-BERT, Streamlit, Librosa, ReportLab</i>	High
4	Transactions processing	<i>Upload/process audio and generate report</i>	High
5	Reporting	<i>Generate downloadable PDF report</i>	High
6	Business rules, etc.	<i>Compare explanation with reference concept</i>	High

7	Compliance to laws or regulations	<i>Protect uploaded user audio</i>	Medium
8	<i>Other</i>	<i>Display semantic similarity, filler word ratio, pause ratio, confidence (energy) score, and final understanding score</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>Audio processed quickly</i>	<i>< 60 seconds for a 30-60 second audio clip (excluding first-run model download)</i>
2	Scalability	<i>Multiple student sessions</i>	<i>100 users</i>
3	Security & Data Privacy	<i>Secure handling of uploaded audio</i>	<i>HTTPS & local processing</i>
4	Reliability & Availability	<i>Stable application</i>	<i>99% availability</i>
5	Usability & Accessibility	<i>Easy-to-use interface</i>	<i>Beginner-friendly</i>
6	<i>Other</i>	<i>Maintainable Python code</i>	<i>Modular architecture</i>